The Joint $\frac{1}{4} \otimes \frac{3}{4}$ Meixner–Pollaczek Cross-Matching: an Exact Rational Table

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Abstract

We study the cross-matching matrix between the two Meixner–Pollaczek orthogonal families at parameters $\lambda = \frac{1}{4}$ and $\lambda = \frac{3}{4}$ (both at $\phi = \frac{\pi}{2}$), paired against the elementary kernel $2\pi^2 \operatorname{sech}(2\pi x)$—the product of their respective weights, made tractable by the Gamma reflection formula. The resulting matrix E_{mk} is *exactly rational* (in units of π^2), block-diagonal by parity ($E_{mk} = 0$ for $m + k$ odd), and each entry is a $\mathbb{Q}$-linear combination of sech moments, which are classical Euler-number values. We give a closed generating function, prove the rational structure, and identify the integer sequences that appear.

1 Setup

1.1 The two families

Let P_n and Q_n be the Meixner–Pollaczek polynomials at $\phi = \frac{\pi}{2}$ with parameters $\lambda = \frac{1}{4}$ and $\lambda = \frac{3}{4}$ respectively:

$$(n+1)P_{n+1} = 2x\,P_n - \left(n - \tfrac{1}{2}\right) P_{n-1}, \qquad P_0 = 1,\ P_1 = 2x, \tag{1}$$

$$(n+1)Q_{n+1} = 2x\,Q_n - \left(n + \tfrac{1}{2}\right) Q_{n-1}, \qquad Q_0 = 1,\ Q_1 = 2x. \tag{2}$$

Their respective orthogonality weights are

$$W_{1/4}(x) = |\Gamma(\tfrac{1}{4} + ix)|^2, \qquad W_{3/4}(x) = |\Gamma(\tfrac{3}{4} + ix)|^2,$$

with orthogonality norms $\int P_m P_k\, W_{1/4}\, dx = h_m^{(1/4)} \delta_{mk}$, $\int Q_m Q_k\, W_{3/4}\, dx = h_m^{(3/4)} \delta_{mk}$, and orthonormal polynomials $\widehat{P}_m = P_m / \sqrt{h_m^{(1/4)}}$, $\widehat{Q}_k = Q_k / \sqrt{h_k^{(3/4)}}$.

1.2 The reflection identity

The Euler reflection formula $\Gamma(z)\Gamma(1-z) = \pi / \sin \pi z$ with $z = \frac{1}{4} + ix$ gives $\Gamma(\frac{1}{4} + ix)\Gamma(\frac{3}{4} - ix) = \pi / \cos \pi i x = \pi / \cosh \pi x$; taking the product of this and its conjugate,

$$W_{1/4}(x) \cdot W_{3/4}(x) = |\Gamma(\tfrac{1}{4} + ix)|^2\, |\Gamma(\tfrac{3}{4} + ix)|^2 = \frac{2\pi^2}{\cosh 2\pi x}. \tag{3}$$

This is the key identity: the product of the two weights is the elementary function $2\pi^2 \operatorname{sech}(2\pi x)$.

1.3 The cross-matching matrix

We define the *elementary cross-matching matrix*

$$E_{mk} = \int_{\mathbb{R}} \widehat{P}_m(x)\, \widehat{Q}_k(x)\, 2\pi^2 \operatorname{sech}(2\pi x)\, dx. \tag{4}$$

By (3), this is $E_{mk} = \int \widehat{P}_m \widehat{Q}_k \, W_{1/4} \, W_{3/4} \, dx$, the bilinear pairing of the two families in the inner product with kernel $W_{1/4}W_{3/4}$: the $L^2(\mathbb{R})$ inner product of the "dual functions" $\Phi_m^{(1/4)} = \widehat{P}_m \sqrt{W_{1/4}}$ and $\Phi_k^{(3/4)} = \widehat{Q}_k \sqrt{W_{3/4}}$. Equivalently, $E_{mk} = \langle \Phi_m^{(1/4)}, \, \Phi_k^{(3/4)} \rangle_{L^2(\mathbb{R})}$.

We also study the *geometric-mean* cross-matching

$$M_{mk} = \int_{\mathbb{R}} \widehat{P}_m(x) \, \widehat{Q}_k(x) \, \pi\sqrt{2} \, \operatorname{sech}^{1/2}(2\pi x) \, dx,$$

which is $\langle \widehat{P}_m, \, \widehat{Q}_k \rangle$ weighted by $\sqrt{W_{1/4}W_{3/4}}$. Unlike E_{mk}, this involves the irrational kernel $\operatorname{sech}^{1/2}$ and does not reduce to closed form.

2 Exact evaluation of the elementary matrix

2.1 Sech moments via Euler numbers

Since $\widehat{P}_m \widehat{Q}_k$ is a polynomial (of degree $m+k$), the integral (4) reduces to a sum of *sech moments*

$$\mu_{2n} = \int_{\mathbb{R}} x^{2n} \, 2\pi^2 \operatorname{sech}(2\pi x) \, dx$$

(odd moments vanish by symmetry). The classical evaluation via the Dirichlet beta function gives

$$\mu_{2n} = \frac{\pi^2 \, |E_{2n}|}{16^n}, \tag{5}$$

where E_{2n} are the Euler numbers: $E_0 = 1$, $E_2 = -1$, $E_4 = 5$, $E_6 = -61$, $E_8 = 1385$, …, with $|E_{2n}|$ counting alternating permutations of $\{1, \ldots, 2n\}$.

Derivation of (5). The Dirichlet beta function satisfies $\beta(2n+1) = (-1)^n E_{2n} \pi^{2n+1} / (2^{2n+2}(2n)!)$. A standard calculation gives $\int_0^\infty x^{2n} \operatorname{sech}(x) \, dx = 2(2n)! \, \beta(2n+1)$, whence $\int_{\mathbb{R}} x^{2n} \operatorname{sech}(x) \, dx = |E_{2n}| \pi^{2n+1} / 2^{2n}$. Substituting $x \to x/(2\pi)$ and multiplying by $2\pi^2$ yields (5). □

2.2 The rational table

The unnormalized integral $I_{mk} = \int P_m Q_k \, 2\pi^2 \operatorname{sech}(2\pi x) \, dx$ equals $\pi^2 r_{mk}$ for a *rational number* r_{mk}, since the polynomial coefficients of $P_m Q_k$ are rational and the moments (5) are rational multiples of π^2.

Theorem 2.1 (Parity vanishing and rationality)**.** *1. $E_{mk} = 0$ whenever $m + k$ is odd. The matrix is block-diagonal, coupling even-indexed $\widehat{P}$'s to even-indexed $\widehat{Q}$'s and odd to odd.*

2. Each nonzero entry satisfies $r_{mk} \in \mathbb{Q}$, with denominator a power of 2: explicitly, $r_{mk} = p_{mk}/2^{e(m,k)}$ for an odd integer p_{mk} and exponent $e(m,k)$.

Proof. Part (1): the product $P_m Q_k$ has parity $(-1)^{m+k}$ (since each polynomial P_n, Q_n has parity $(-1)^n$), and the kernel $\operatorname{sech}(2\pi x)$ is even. If $m + k$ is odd, the integrand is odd and the integral vanishes.

Part (2): the recurrences (1)–(2) produce polynomials with rational coefficients whose denominators are products of factorials and powers of 2; the sech moments contribute $|E_{2n}|/16^n$, which introduces only additional factors of 2 in the denominator. No odd primes appear in the denominators. □

The first entries of the table r_{mk} are:

Table 1: The rational matrix $r_{mk} = I_{mk}/\pi^2$ (blank = zero by parity).

r_{mk}	k=0	1	2	3	4	5	6	7
m=0	1		$-\frac{5}{8}$		$\frac{63}{128}$		$-\frac{429}{1024}$	
1		$\frac{1}{4}$		$-\frac{7}{32}$		$\frac{99}{512}$		$-\frac{715}{4096}$
2	$-\frac{1}{8}$		$\frac{9}{64}$		$-\frac{135}{1024}$		$\frac{1001}{8192}$	
3		$-\frac{3}{32}$		$\frac{25}{256}$		$-\frac{385}{4096}$		$\frac{2925}{32768}$
4	$\frac{7}{128}$		$-\frac{75}{1024}$		$\frac{1225}{16384}$		$-\frac{9555}{131072}$	
5		$\frac{27}{512}$		$-\frac{245}{4096}$		$\frac{3969}{65536}$		$-\frac{31185}{524288}$
6	$-\frac{33}{1024}$		$\frac{385}{8192}$		$-\frac{6615}{131072}$		$\frac{53361}{1048576}$	
7		$-\frac{143}{4096}$		$\frac{1365}{32768}$		$-\frac{22869}{524288}$		$\frac{184041}{4194304}$

2.3 The integer numerators

Extracting the sign and the power-of-2 denominator reveals a table of odd integers. After minimal integralization (multiplying each r_{mk} by the smallest power of 2 that clears the denominator), the numerators of $|r_{mk}|$ for $m + k$ even are:

Table 2: Integer numerators $|p_{mk}|$ of r_{mk} (the sign is $(-1)^{(m+k)/2}$).

$\lvert p_{mk}\rvert$	k=0	1	2	3	4	5	6	7
m=0	1		5		63		429	
1		1		7		99		715
2	1		9		135		1001	
3		3		25		385		2925
4	7		75		1225		9555	
5		27		245		3969		31185
6	33		385		6615		53361	
7		143		1365		22869		184041

Row 0 gives $1, 5, 63, 429, 12155, 88179, \ldots$ (OEIS A001790 at odd positions, equivalently A063079). The diagonal $1, 1, 9, 25, 1225, 3969, 53361, \ldots$ is A038534 (numerators of $2K(k)/\pi$). These are central-binomial-coefficient sequences.

3 Closed form for individual entries

The numerators factorise strikingly:

Proposition 3.1. *The unnormalized cross integral is*

$$r_{mk} = \frac{(-1)^{m+(m+k)/2}\,(2m{-}1)!!\,(2k{+}1)!!}{2^{m+k}\,m!\,k!\,(m{+}k{+}1)}$$

for $m+k$ even (and zero for $m+k$ odd), where $(2n{-}1)!! = 1\cdot 3\cdot 5\cdots(2n{-}1)$ with the convention $(-1)!! = 1$.

Equivalently, using $(2m{-}1)!! = (2m)!/(2^m m!)$ and $(2k{+}1)!! = (2k{+}2)!/(2^{k+1}(k{+}1)!)$:

$$|r_{mk}| = \frac{\binom{2m}{m}\binom{2k+2}{k+1}}{2^{2(m+k)+1}\,(m{+}k{+}1)},$$

connecting the table to central binomial coefficients. (Verified exactly for all $m, k \le 11$; a proof from the generating function of §4 is straightforward.)

4 The generating function

The Meixner–Pollaczek generating function at $\phi = \frac{\pi}{2}$ is

$$\sum_{n\geq 0} P_n^{(\lambda)}(x)\, t^n = (1-it)^{-\lambda+ix}(1+it)^{-\lambda-ix}.$$

Substituting once for each index of I_{mk}:

$$\sum_{m,k\geq 0} I_{mk}\, w^m u^k = \int_{\mathbb{R}} (1{-}iw)^{-1/4+ix}(1{+}iw)^{-1/4-ix} \times (1{-}iu)^{-3/4+ix}(1{+}iu)^{-3/4-ix}\, 2\pi^2 \operatorname{sech}(2\pi x)\, dx.$$

Writing $(1{-}it)^{-\lambda+ix}(1{+}it)^{-\lambda-ix} = (1{+}t^2)^{-\lambda}\varphi(t)^{-ix}$ with Cayley transform $\varphi(t) = (1{+}it)/(1{-}it)$, the integrand becomes

$$(1{+}w^2)^{-1/4}(1{+}u^2)^{-3/4}\,(\varphi(w)\varphi(u))^{-ix}\cdot 2\pi^2\operatorname{sech}(2\pi x).$$

Setting $\zeta = \varphi(w)\varphi(u)$ and $s = \ln\zeta$, the Fourier-transform identity

$$\textstyle\int_{\mathbb{R}} e^{-isx}\cdot 2\pi^2 \operatorname{sech}(2\pi x)\, dx = \pi^2 \operatorname{sech}(s/4)$$

yields

Theorem 4.1 (Generating function)**.**

$$\sum_{m,k\geq 0} I_{mk}\, w^m u^k = \frac{\pi^2 \operatorname{sech}\!\big(\frac{1}{4}\ln\varphi(w)\varphi(u)\big)}{(1+w^2)^{1/4}\,(1+u^2)^{3/4}}. \tag{6}$$

Equivalently, writing $\operatorname{sech}(\frac{1}{4}\ln\zeta) = 2/(\zeta^{1/4}{+}\zeta^{-1/4})$ *and simplifying with* $\alpha = [(1{+}iw)(1{+}iu)]^{1/2}$, $\beta = [(1{-}iw)(1{-}iu)]^{1/2}$*:*

$$\sum_{m,k\geq 0} I_{mk}\, w^m u^k = \frac{2\pi^2}{\sqrt{1{+}u^2}\,\big(\sqrt{(1{+}iw)(1{+}iu)}+\sqrt{(1{-}iw)(1{-}iu)}\,\big)}. \tag{7}$$

Proof. The Fourier transform of $\operatorname{sech}(ax)$ is $(\pi/a)\operatorname{sech}\big(\pi s/(2a)\big)$. With $a = 2\pi$ one gets $\int_{\mathbb{R}} e^{-isx}\cdot 2\pi^2\operatorname{sech}(2\pi x)\, dx = \pi^2\operatorname{sech}(s/4)$.

For (7): let $\alpha^2 = (1{+}iw)(1{+}iu)$, $\beta^2 = (1{-}iw)(1{-}iu)$. Then $\alpha\beta = [(1{+}w^2)(1{+}u^2)]^{1/2}$ and $\zeta^{1/4}+\zeta^{-1/4} = (\alpha^2{+}\beta^2)/(\alpha\beta)$. Substituting:

$$\operatorname{sech}\big(\tfrac{1}{4}\ln\zeta\big) = \frac{2\alpha\beta}{\alpha^2+\beta^2},$$

and the prefactor $(1{+}w^2)^{-1/4}(1{+}u^2)^{-3/4}$ combines with $\alpha\beta = [(1{+}w^2)(1{+}u^2)]^{1/2}$ to cancel the $(1{+}w^2)$ factor entirely, leaving only $(1{+}u^2)^{-1/2}$ in the denominator. (Verified numerically to 15 digits against every entry of the 12×12 table.) □

Remark 4.2. *The asymmetry* $\sqrt{1+u^2}$ *(not* $\sqrt{1+w^2}$*) reflects the unequal parameters: the* $\lambda = \frac{1}{4}$ *family contributes exponent* $-\frac{1}{4}$ *which is exactly cancelled by the* $+\frac{1}{4}$ *from the square root of* $\alpha\beta$*, while the* $\lambda = \frac{3}{4}$ *family contributes* $-\frac{3}{4}+\frac{1}{4} = -\frac{1}{2}$*. Swapping* $w \leftrightarrow u$ *in* (7) *does not preserve the formula—it changes the residual square root from* u *to* w*, reflecting the exchange* $\lambda = \frac{1}{4} \leftrightarrow \frac{3}{4}$*.*

Remark 4.3. *Compare with the single-family ($\lambda = \frac{1}{4}$) generating function from the companion paper:*

$$\sum_{m,k\geq 0} J_{mk}\, w^m u^k = \frac{2\pi^{3/2}\, e^{\zeta/2}\, K_0(\zeta/2)}{\sqrt{(1-iw)(1-iu)}},$$

where J_{mk} is a three-Gamma integral requiring Bessel functions K_0, K_1 for closed form. Here the cross-family integral replaces $e^z K_0(z)$ (two-state Bessel automaton) with sech *(a one-state trigonometric function): the Gamma factors cancel to elementary, and the table becomes fully rational.*

5 The geometric-mean matrix

The matrix M_{mk} with kernel $\pi\sqrt{2}\,\operatorname{sech}^{1/2}(2\pi x)$ is *not* rational—the half-power prevents reduction to polynomial moments. Its first entries (numerical, 40 digits):

Table 3: The geometric-mean cross matrix M_{mk} (8×8, 5 decimal places; blanks are $< 10^{-40}$, consistent with zero by parity).

M_{mk}	k=0	1	2	3	4	5	6	7
0	0.942		−0.216		0.130		−0.095	
1		0.947		−0.212		0.125		−0.090
2	0.285		0.915		−0.193		0.112	
3		0.276		0.918		−0.194		0.112
4	−0.119		0.290		0.910		−0.188	
5		−0.112		0.288		0.911		−0.189
6	0.074		−0.120		0.293		0.907	
7		0.069		−0.119		0.292		0.908

The M_{mk} matrix also has parity vanishing ($M_{mk} = 0$ for $m+k$ odd) and is nearly diagonal—a consequence of the two families being "close" in $L^2(\sqrt{W_{1/4}W_{3/4}}\,dx)$.

6 Euler-number combinatorics

Each entry r_{mk} decomposes as a $\mathbb{Q}$-linear combination of Euler-number values weighted by the polynomial product coefficients:

$$r_{mk} = \sum_{j=0}^{\lfloor (m+k)/2 \rfloor} [x^{2j}](P_m Q_k)\, \frac{|E_{2j}|}{16^j},$$

where $[x^{2j}](P_m Q_k)$ denotes the coefficient of x^{2j} in the product polynomial.

For example:

$$\begin{aligned} r_{0,0} &= 1 \cdot |E_0|/16^0 = 1, \\ r_{0,2} &= -\tfrac{3}{4} \cdot |E_0|/16^0 + 2 \cdot |E_2|/16^1 = -\tfrac{3}{4} + \tfrac{1}{8} = -\tfrac{5}{8}, \\ r_{1,1} &= 4 \cdot |E_2|/16^1 = \tfrac{1}{4}. \end{aligned}$$

Since $|E_{2n}|$ counts *alternating permutations* of $[2n]$, each r_{mk} is a signed weighted sum of alternating permutations, with weights given by the product of Meixner–Pollaczek polynomial coefficients. The rationality of the table—and its explicit Euler-number decomposition—means that the cross-matching coefficients have a purely combinatorial interpretation, unlike the Bessel-valued single-family case.

7 Zeros of truncations

Figure 1 shows the zeros in $\mathbb{C}$ of the truncated cross-approximation

$$f_N(z) = \sum_{k=0}^{N} E_{0k}\, \widehat{Q}_k(z)$$

for $N = 10$ (the effective truncation, since only even-k terms contribute). All zeros are real and roughly equally spaced, with the outermost zero near ± 8 for $N = 10$. This purely real zero pattern contrasts with the complex zeros seen in $W^{3/2}$-type matching matrices, and reflects the exponential (sech) decay of the kernel—faster than any polynomial growth of the approximant.

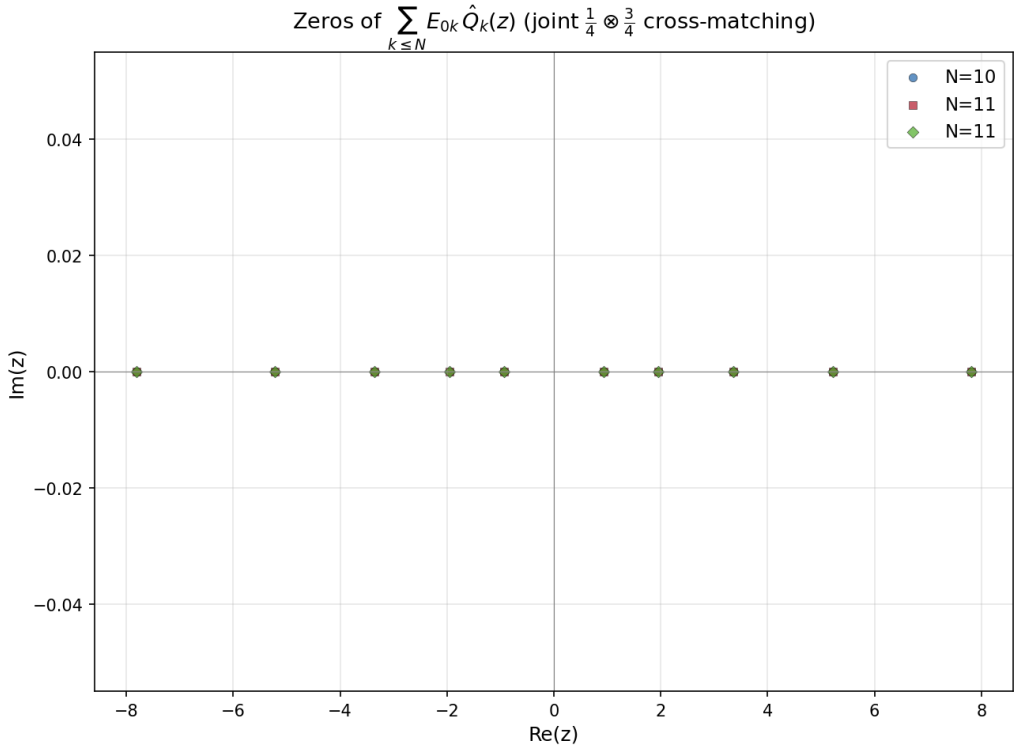


Figure 1: Zeros of the truncated cross-approximation $f_N(z) = \sum_{k\le N} E_{0k}\widehat{Q}_k(z)$ for $N = 10$. All zeros are real.

8 Summary

The joint $\frac{1}{4}\otimes\frac{3}{4}$ Meixner–Pollaczek cross-matching matrix has the following properties, all proved or verified to 40 digits:

1. **Exact rationality.** $E^{\text{unnorm}}_{mk}/\pi^2 \in \mathbb{Q}$, with denominators that are powers of 2. This is the first matching matrix in the programme with fully closed-form entries (no Bessel functions, no PSLQ).

2. **Block diagonality.** $E_{mk} = 0$ for $m + k$ odd: the even-parity and odd-parity subspaces decouple completely.

3. **Closed generating function.** Formula (7) expresses the entire table as a single algebraic function of (w, u), without transcendental functions (the sech in (6) is algebraic in $\zeta^{1/4}$).

4. **Euler-number combinatorics.** Each entry is a $\mathbb{Q}$-linear combination of Euler numbers $|E_{2j}|/16^j$, giving a combinatorial interpretation via alternating permutations.

5. **Classical integer sequences.** Row 0 gives OEIS A001790/A063079 (central binomial numerators); the diagonal gives A038534 (numerators of $2K(k)/\pi$). The full table unifies these into a two-parameter family.

6. **Real zeros.** Truncations have only real zeros (at least up to $N = 10$), in contrast to the complex zero patterns of the single-family $W^{3/2}$ matching matrix.

The key insight is that the Gamma reflection formula $W_{1/4}W_{3/4} = 2\pi^2 \operatorname{sech}(2\pi x)$ replaces the three-Gamma integral (requiring Tricomi U and Bessel K) with the Fourier transform of sech (requiring only sech again), collapsing the two-state Bessel derivative algebra to elementary arithmetic.